

Phishing Detection & Awareness Report

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1. Introduction



Phishing is a common cybersecurity attack where attackers attempt to trick individuals into revealing sensitive information such as passwords, banking details, or personal data.

These attacks are typically carried out through fraudulent emails that appear to come from legitimate sources. Phishing attacks are dangerous because they exploit human trust and can lead to financial loss, identity theft, and unauthorized access to systems.





2. Objective

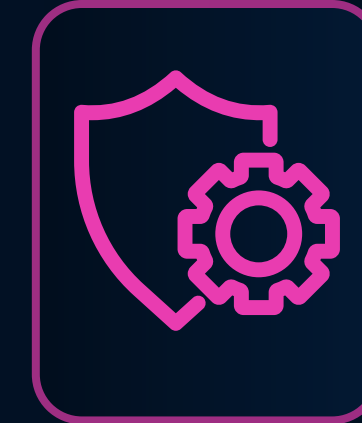
The objective of this project is to:

- Identify phishing characteristics in email samples
- Classify emails as Safe, Suspicious, or Phishing
- Explain phishing techniques in simple, user-friendly language
- Provide awareness and prevention guidelines for users

3. Methodology

The following approach was used to analyze the email samples:

- Verifying the sender's email address for authenticity
- Examining subject lines for urgency or threatening language
- Checking embedded links for suspicious or fake domains
- Reviewing email content for tone, grammar, and intent
- Identifying requests for sensitive information or attachments



Email Analysis

Email 1

Subject: Urgent: Your Account Will Be Suspended

Sender: support@paypal.com

Classification: Phishing

Indicators:

- Spoofed sender address (paypal.com instead of paypal.com)
- Use of urgent and threatening language
- Suspicious link (paypal-secure-login.xyz)
- Generic greeting ("Dear Customer")
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Explanation:

This email attempts to create panic and urgency to trick the user into clicking a malicious link. The sender address is slightly altered to resemble a legitimate company, which is a common phishing technique used by attackers.



Email Analysis

Email 2

Subject: Important: Verify Your Bank Account

Sender: security@bank-update.com

Classification: Phishing

Indicators:

- Fake domain not associated with a legitimate bank
- Request for sensitive financial information
- Threatening tone (account closure warning)
- Mention of attachments that may contain malware

Explanation:

This email is designed to steal banking information by pressuring the user. Attackers use fear tactics to make users act quickly without verifying the authenticity of the message.



Email **Analysis**

Email 3

Subject: Important: Verify Your Bank Account

Sender: security@bank-update.com

Classification: Phishing

Indicators:

- Fake domain not associated with a legitimate bank
- Request for sensitive financial information
- Threatening tone (account closure warning)
- Mention of attachments that may contain malware

Explanation:

This email is designed to steal banking information by pressuring the user. Attackers use fear tactics to make users act quickly without verifying the authenticity of the message.



5. Common Phishing Techniques

Attackers commonly use the following techniques:



- Email Spoofing: Creating fake sender addresses that appear legitimate



- Urgency and Fear Tactics: Pressuring users to act immediately
- Fake Websites: Redirecting users to malicious websites that mimic real ones



- Link Manipulation: Hiding malicious URLs behind trusted-looking text



- Attachment-Based Attacks: Sending infected files to install malware

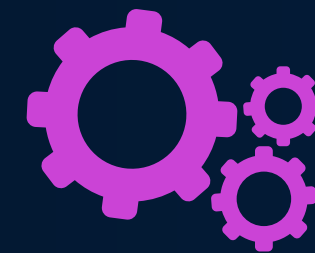


6. Prevention & Awareness Guidelines

Effects

To protect against phishing attacks, users should follow these best practices:

- Always verify the sender's email address carefully
- Avoid clicking on suspicious or unknown links
- Hover over links to check the actual URL before clicking
- Do not download unexpected attachments
- Never share sensitive information via email
- Enable Two-Factor Authentication (2FA)
- Use updated antivirus and security software
- Report suspicious emails to the appropriate authority or IT team



7. Conclusion



- Phishing attacks continue to be one of the most effective cyber threats due to their reliance on human behavior. By understanding common phishing techniques and practicing safe email habits, individuals can significantly reduce the risk of being targeted.
- User awareness and cautious behavior are essential in preventing phishing attacks and ensuring overall cybersecurity.





**Thank
You!**